OPEN WEATHER API

In [112…

```
import json
data = [{
    'name':"John",
    'age':34,
    'city':"Ahmedabad"
}]
json_string = json.dumps(data,indent=5)
print(json_string)
```


```
[
     {
          "name": "John",
          "age": 34,
          "city": "Ahmedabad"
     }
]
```


1. Geocode API

link : https://openweathermap.org/api/geocoding-api

using Geocode API to obtain coordinates(latitude and longitude) of any city

In [66]:

```
import requests,json

# api_key = "21cd5b4cda0bfc726947cf07664fb9d5"
api_key = input("Enter the API key :")
city_name = input("Enter the city name :")
url = f"https://api.openweathermap.org/geo/1.0/direct?q={city_name}&appid={api_key}"

response = requests.get(url)
# print(response.status_code)

data = response.json()
print(data)
```

```
# print(json.dumps(data,indent=5))

latitude = data[0]['lat']
print("latitude:",latitude)

longitude = data[0]['lon']
print("longitude:",longitude)
```


```
[{'name': 'Ahmedabad', 'local_names': {'kn': 'ಅಹ್ಮದಾಬಾದ್', 'ml': 'അഹമ്മദാബാദ്', 'tt': 'әхмәтабад', 'feature_nam
e': 'Ahmedabad', 'mr': 'अहमदाबाद', 'eo': 'Ahmadabado', 'te': 'అహ్మదాబాద్', 'ru': 'Ахмадабад', 'pa': 'ਅਹਿਮਦਾਬਾਦ', 'b
n': 'আহমেদাবাদ', 'he': 'אחמדאבאד', 'ta': 'அகமதாபாத்', 'ar': 'أحمد آباد', 'pl': 'Ahmadabad', 'uk': 'Ахмедабад', 'oc':
'Ahmadabad', 'ur': 'احمد آباد', 'ja': 'アフマダーバード', 'de': 'Ahmedabad', 'ascii': 'Ahmedabad', 'hi': 'अहमदाबाद', 'cs':
'Ahmadábád', 'ne': 'अहमदाबाद', 'or': 'ଅହମଦାବାଦ', 'zh': '艾哈迈达巴德', 'en': 'Ahmedabad', 'gu': 'અમદાવાદ'}, 'lat': 23.
0215374, 'lon': 72.5800568, 'country': 'IN', 'state': 'Gujarat'}]
```


```
latitude: 23.0215374
longitude: 72.5800568
```

2. Current Weather API

link : https://openweathermap.org/api/current

using Current Weather API to obtain current weather details of any city

In [118…

```
import requests,json

api_key = "21cd5b4cda0bfc726947cf07664fb9d5"
# api_key = input("Enter the API key :")

url = f"https://api.openweathermap.org/data/2.5/weather?lat={latitude}&lon={longitude}&appid={api_key}"

res = requests.get(url)
# print(res)

data = res.json()
# print(json.dumps(data,indent=5))

weather_desc = data['weather'][0]['description']
# print("Weather Description :",weather_desc)

print("Current Weather : ")
```

```
temperature = data['main']['temp']
print("Temperature : ",temperature)

pressure = data['main']['pressure']
print("Pressure : ",pressure)

humidity = data['main']['humidity']
print("Humidity :",humidity)

wind_speed = data['wind']['speed']
print("Wind Speed : ",wind_speed)

visibility = data['visibility']
print("Visibility : ",visibility)
```

```
Current Weather :
Temperature :  308.4
Pressure :  1004
Humidity : 43
Wind Speed :  4.97
Visibility :  10000
```

3. Air Pollution API

link : https://openweathermap.org/api/air-pollution

using Air Pollution API to obtain current air pollution details of any city

In [103…

```
import requests,json

api_key = "21cd5b4cda0bfc726947cf07664fb9d5"
# api_key = input("Enter the API key :")

quality_indices = {1:'Good',2:'Fair',3:'Moderate',4:'Poor',5:'Very Poor'}

url = f"http://api.openweathermap.org/data/2.5/air_pollution?lat={latitude}&lon={longitude}&appid={api_key}"

res = requests.get(url)

data = res.json()
# print(json.dumps(data,indent=5))
```

```
index = data['list'][0]['main']['aqi']

print("current Air Quality:",index)
print("current Air Quality:",quality_indices[index])
```

```
current Air Quality: 2
current Air Quality: Fair
```

4. Air Pollution API (Forecasting)

link : https://openweathermap.org/api/air-pollution?collection=environmental#forecast

using Air Pollution API to obtain history and air pollution details

In [145…

```
import requests,json

api_key = "21cd5b4cda0bfc726947cf07664fb9d5"
# api_key = input("Enter the API key :")

url = f"http://api.openweathermap.org/data/2.5/air_pollution/forecast?lat={latitude}&lon={longitude}&appid={api_key}'

res = requests.get(url)

data = res.json()
# print(json.dumps(data,indent=5))

# print(data['list'])

D = {'date_time':[],'co':[],'no':[],'no2':[],'o3':[],'so2':[],'pm2_5':[],'pm10':[],'nh3':[]}

for i in data['list']:
    D['date_time'].append(i['dt'])
    D['co'].append(i['components']['co'])
    D['no'].append(i['components']['no'])
    D['no2'].append(i['components']['no2'])
    D['o3'].append(i['components']['o3'])
    D['so2'].append(i['components']['so2'])
    D['pm2_5'].append(i['components']['pm2_5'])
    D['pm10'].append(i['components']['pm10'])
    D['nh3'].append(i['components']['nh3'])
```

```python
import pandas as pd
df = pd.DataFrame(D)
df.head()
```

Out[145…

	date_time	co	no	no2	o3	so2	pm2_5	pm10	nh3
0	1780545600	101.84	0.47	1.88	47.87	1.94	12.83	49.67	7.79
1	1780549200	99.95	0.30	1.39	59.13	1.90	12.53	45.64	6.88
2	1780552800	99.56	0.22	1.12	67.89	1.95	13.20	44.39	6.63
3	1780556400	100.24	0.15	0.89	73.86	1.65	11.82	36.89	5.58
4	1780560000	100.03	0.13	0.77	75.64	1.36	10.08	30.22	4.83